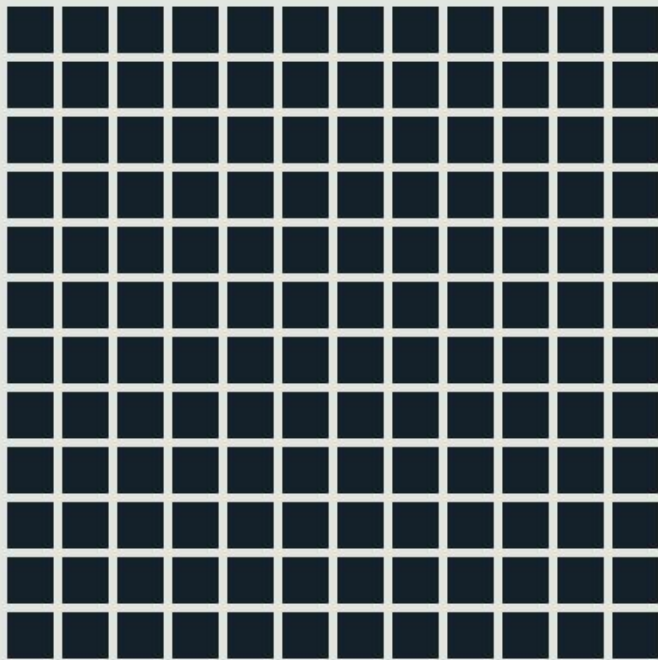


What if the model didn't have to look at everything?

A possible thesis direction on making
Transformer inference dramatically
cheaper.



In dense self-attention, each token can be compared against the rest of the sequence.



n tokens

n^2 comparisons

Double the input and the work quadruples. Every one of those cells gets computed.



The question I want
to investigate:

Can we identify whole
regions of attention that
don't need exact evaluation
— and skip or approximate
them before computing
everything inside?



Dense attention evaluates the full interaction space.

DENSE

compute every interaction → then see what mattered

PROPOSED

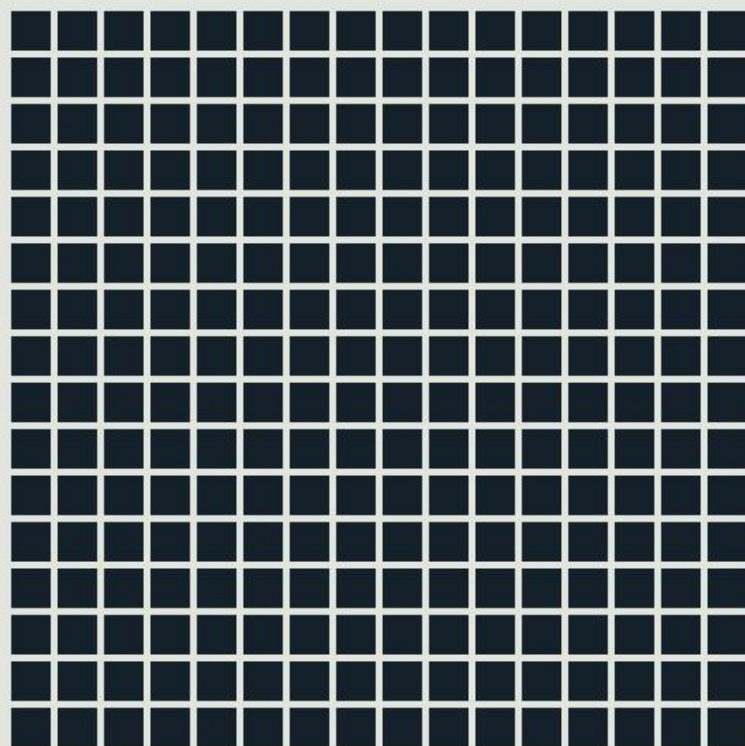
decide per region → skip or approximate → compute less

Decide whether a region can be skipped or approximated before computing the individual interactions inside it.



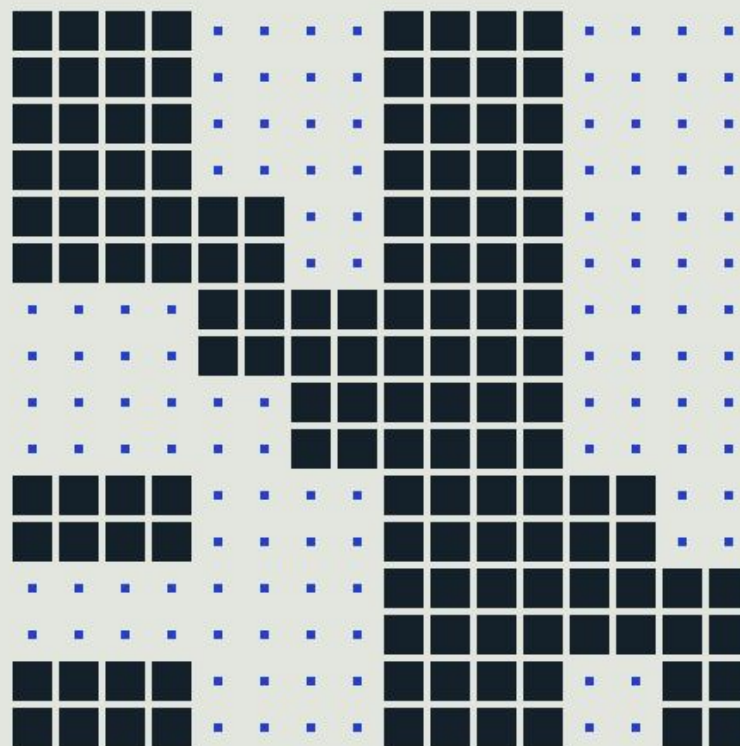
The same layer, two ways

FULL DENSE



all regions evaluated

HIERARCHICALLY REDUCED



fewer regions evaluated

-
- exact interaction region, evaluated
 - eliminated or approximated beforehand



What that could buy

- **Less computation**

fewer operations while preserving acceptable model quality

- **Lower hardware requirements**

making capable local inference more practical

- **Lower energy use**

potentially reducing power use by avoiding unnecessary work

- **More capable AI running locally**

on ordinary computers, privately



It may succeed,
partially succeed,
or fail.

The worst case may still be quadratic. The real question is whether actual workloads carry enough structure to exploit — without losing model quality.

That gets settled by building it and testing it, not by assuming it.